FAIL-CLOSED GATING

INPUT DATA

Target Requirement (Requirement Unit)
req_id: str | type: REQ_TYPE | section: str | text: str

Enclosing SRS Context (SRSContext)
Domain Summary • Domain Glossary • Surrounding Window (±3 Reqs) • Applicable Standards

LAYER 1 Requirement Quality Analysis (Layer1DocumentScorer) Pydantic: Layer1Output

7 ISO/IEC/IEEE 29148 Quality Attributes:

Unambiguity (w=0.2) | Completeness (0.2) | Consistency (0.1) | Conformance (0.1) | Correctness (0.1) | Singularity (0.1) | Verifiability (0.2)

Deterministic Quality Score Calculation (Python Runtime)
$Q_{req} = \Sigma (w_i \cdot S(A_i)) \in [0.0, 10.0]$ | Prevents Hallucinated Math

Candidate Defect Trigger Rule
If $S(A_i) < 8.0$ ➔ Emit candidate Shortcoming(id, attr, excerpt, explanation)

Passes: D_L1 Candidate Defects

LAYER 2 Contextual Defect Reasoning (Layer2DefectReasoner) FSR = |D_discarded| / |D_L1|

Resolves the "Contextual False-Positive Paradox":
Audits candidate defects against surrounding definitions, glossary entries, and companion requirements across the SRS.

DISCARDED DEFECTS False Positives (Resolved in Context)
• Term explicitly defined in Glossary or Standards
• Bound established by neighboring requirement • Suppressed from Generation

RETAINED DEFECTS True Quality Vulnerabilities
• Genuine specification ambiguity, missing bounds, or untestable clauses
➔ Only Retained Defects proceed downstream to Layer 3

Terminated (No downstream action)

Passes: D_retained ONLY

LAYER 3 Surgical Requirement Refinement (Layer3SolutionGenerator) Pydantic: Layer3Output

Constrained Surgical Refinement Engine (Zero Wholesale Rewriting):

1. Strict Scope Lock
Preserve domain entities, system names & intent

2. EARS Syntax
Ubiquitous ("shall"), Event/State-driven rules

3. Defect-Targeted
Resolves ONLY defect IDs present in D_retained

4. Traceable Audit
Emits changes_made & intent_preservation_rationale

Passes: Proposed Refinement + D_retained

LAYER 4 Independent Solution Validation (Layer4SolutionValidator) Pydantic: Layer4Output

Fail-Closed Adversarial Critic (Conjunctive Audit Function: $V = \bigwedge c_j$):

1. Defect Resolution
All D_retained fixed?

2. No New Ambiguity
Zero vague terms added?

3. No Unwarranted Assumptions
No fabricated features?

4. Intent Preservation
Original logic retained?

5. Conformance
IEEE 29148 modal syntax?

Gating Rule: If ANY check fails (False) ➔ pass_validation = False. Proposal is flagged with explicit critic observation.

Passes: Validated Audit Payload

LAYER 5 Human Decision Management (Layer5HumanDecisionManager) Pydantic: Layer5Output

Word-Level Diff Engine
Punctuation-preserving inline deletions & insertions

Complete Chain of Custody
Aggregates rationales from L1 score, L2 suppression trace & L4 validation audit

Actionable Recommendation
Auto-Accept / Stakeholder Review / Manual Override Trigger

OUTPUT ARTIFACT: PipelineExecutionResult status: "completed" | "error" • execution_time_seconds: float • all layer payloads encapsulated